Unit NBIoT2

SKU:U111-B













Description

Unit NBIoT2 is an NB-IoT wireless communication module suitable for global Cat-NB frequency bands. It features the built-in SIM7028 communication module, uses serial communication (controlled via AT command set), and integrates an SMA external antenna interface, significantly improving the communication quality and signal stability of the device. The SIM7028 module supports multi-band NB-IoT network connectivity, ensuring wide applicability worldwide. Designed for low-latency, low-throughput applications, the SIM7028 module has excellent power management capabilities, making it suitable for long-term operation devices. It is ideal for IoT applications such as smart metering, remote control, asset tracking, remote monitoring, telemedicine, and shared bikes.

Features

- Global version/multi-band support
- Strong signal access capability
- AT command control
- SIM card type: Nano
- External antenna: SMA antenna
- Serial communication: UART 115200bps
- Cat-NB frequency bands: B1/B2/B3/B4/B5/B8/B12/B13/B14/B17/B18/B19/B20/B25/B26/B28/B66/B70/B85
- Data transfer rate (kbps): 127(DL)/159(UL)
- Network protocols: TCP/UDP/HTTP/TLS*/DTLS*/DNS/NTP/PING/LWM2M/COAP*/MQTT/MQTTS/SSL
- Certifications:

- SRRC*/NAL*/CCC*
- CE-RED*/GCF*
- Deutsche Telekom*/Vodafone*/Telefonica*
- RoHS*/REACH*
- China Telecom*
- China Mobile*
- China Unicom*

Includes

- 1 x Unit NB-IoT2
- 1 x SMA antenna
- 1 x Hex wrench
- 1 x HY2.0-4P Grove cable (20cm)

Applications

- Remote control
- Asset tracking
- Remote monitoring
- Shared bikes

Specifications

Specification	Parameter
Communication module	SIM7028
Supported Cat-NB bands	B1/B2/B3/B4/B5/B8/B12/B13/B14/B17/B18/B19/B20/B25/B26/B28/B66/B70/B85
Network protocols	TCP/UDP/HTTP/TLS*/DTLS*/DNS/NTP/PING/LWM2M/COAP*/MQTT/MQTTS/SSL
Operator certifications	Deutsche Telekom/Vodafone/Telefonica/China Telecom/China Mobile/China Unicom
Communication method	UART 115200bps
Power consumption test	Standby current: DC 5V/3.1mA Network current: DC 5V/36mA
Product Size	71.5 x 24 x 8mm
Package Size	136 x 92 x 13mm
Product weight	12.4g
Gross Weight	25.5g

Certifications

Radio and Communication Equipment Certifications

- **SRRC**, **NAL**, **CE-RED**, **GCF**, **CCC**

Safety and Environmental Certifications

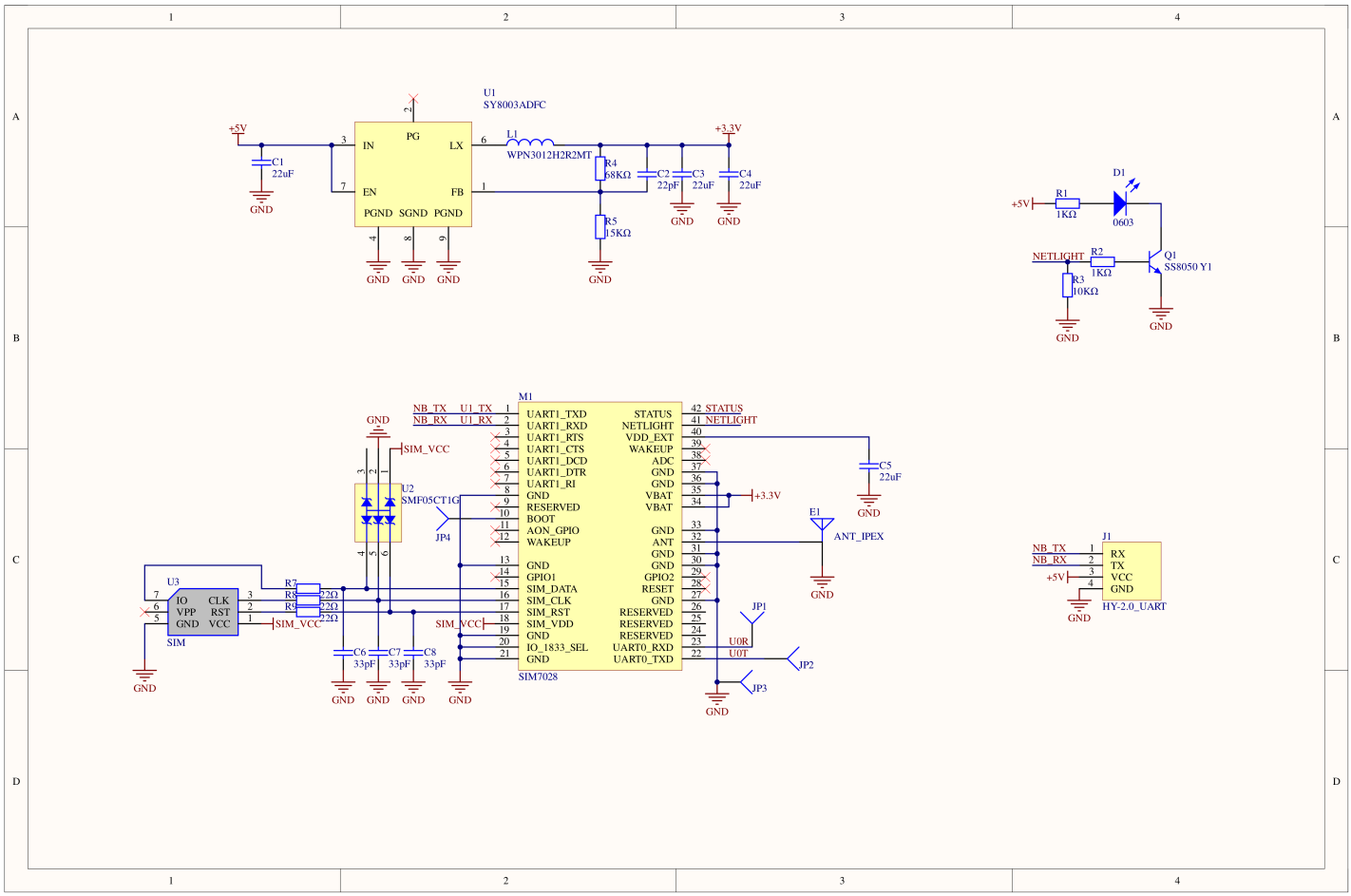
- **CCC**, **RoHS**, **REACH**

Operator Certifications

- Deutsche Telekom, Vodafone, Telefonica, China Telecom, China Mobile, China Unicom

Schematics

- Unit NBIoT2 Schematics PDF

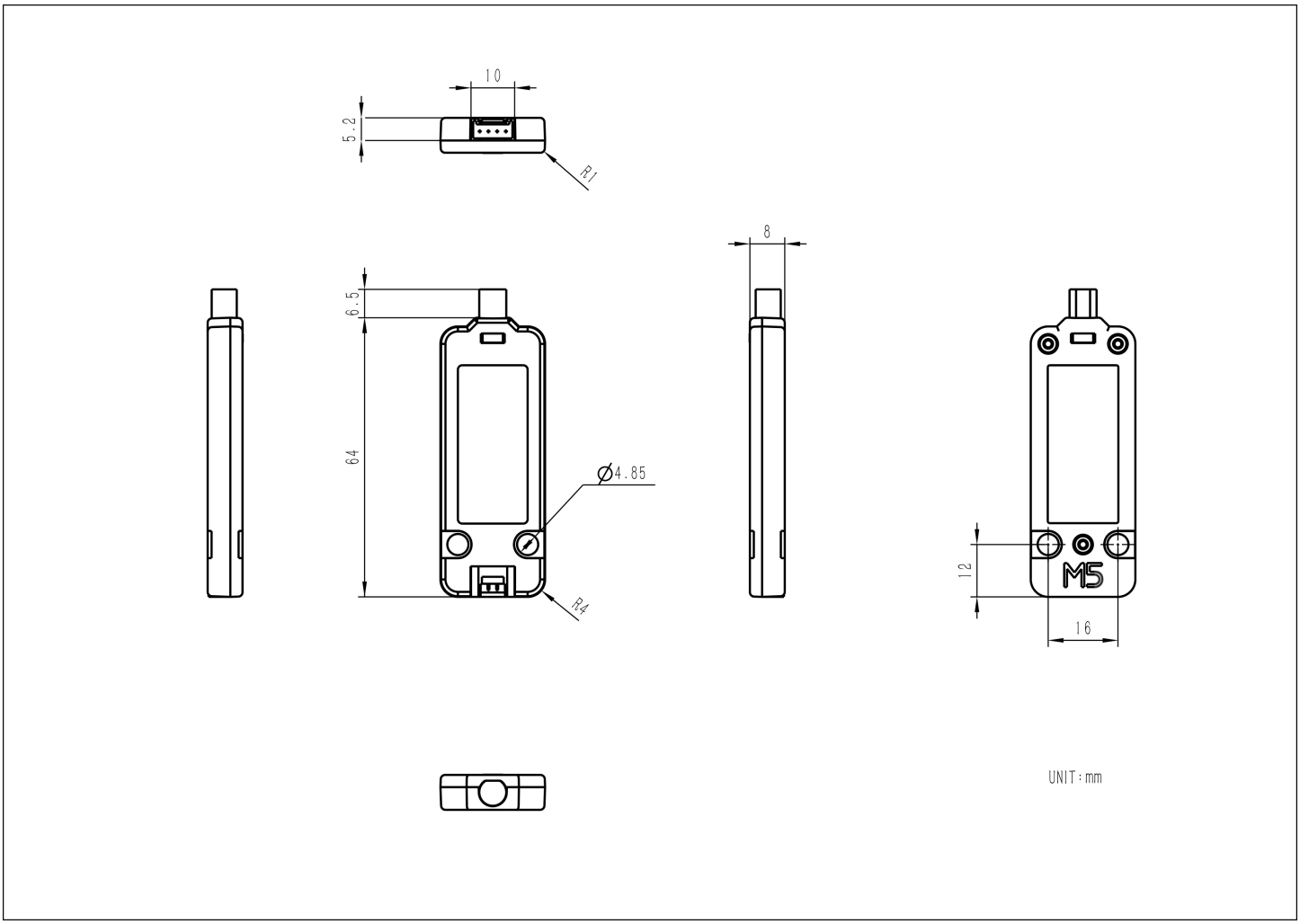


PinMap

Unit NBIoT2

HY2.0-4P	Black	Red	Yellow	White
PORT.C	GND	5V	UART_RX	UART_TX

Model Size



Structure

- Unit NB-IoT2 Structure Files

Datasheets

- SIM7028_Specification
- SIM7028 Series_Network Searching_Application Note_V1.00
- SIM7028 Series_Low Power Mode_Application Note_V1.01
- SIM7028_NCC_Certificate_20230421
- SIM7028 Series_TCPIP_Application Note_V1.04
- SIM7028 Series_SSL_Application Note_V1.00
- SIM7028 Series_MQTT(S)_Application Note_V1.03
- SIM7028 Series_LwM2M_Application Note_V1.02
- SIM7028 Series_HTTP(S)_Application Note_V1.04
- SIM7028 Series_FOTA_Application Note_V1.00
- SIM7028 Series_COAP_Application Note_V1.00
- NB-EVB_User_Guide_Manual_V1.00
- RSU062-EE-EUT Photo-SIM7028
- RSU062-E6-CE EMC Test Report
- RSU062-E5-3GPP (LTE Band 5) Test Report

Softwares

Arduino

- Unit NBIoT2 Arduino Library

Programming Note

PIO users can refer to the following configuration to download the specified TinyGSM library version: v0.12.0-patch-1

Arduino: Download the above library file and manually install it to the specified Arduino library folder.

```
lib_deps =
        https://github.com/m5stack/TinyGSM/archive/refs/tags/v0.12.0-patch-1.zip
```

UiFlow2

- Unit NB-IoT2 UiFlow2 Docs

Protocol

- SIM7028 Series_AT Command Manual_V1.00

Video

U111-B NB-IoT2 UNIT 视频.mp4

Product Comparison

Product	SKU	Module	Cat-NB Frequency Bands	Data Transfer Rate	Communication Protocols
Unit NB-IoT	U111	SIM7020G	B1/B2/B3/B4/B5/B8/B12/B13/B17/B18/B19/B20/B25/B26/B28/B66/B70/B71/B85	126 (DL) / 150 (UL)	TCP/UDP/HTTP/HTTPS/TLS*/DTLS*/DNS/NTP/PING/LWM2M/COAP*/MQTT/MQTTS/SSL
Unit NB-IoT2	U111-B	SIM7028	B1/B2/B3/B4/B5/B8/B12/B13/B14/B17/B18/B19/B20/B25/B26/B28/B66/B70/B85	127 (DL) / 159 (UL)	TCP/UDP/HTTP/HTTPS/TLS*/DTLS*/DNS/NTP/PING/LWM2M/COAP*/MQTT/MQTTS/SSL

Comparison Summary

The main differences between SIM7028 and SIM7020G are in frequency band support and data transfer rates. SIM7028 supports more frequency bands and slightly higher data transfer rates, making it suitable for more markets and higher bandwidth applications. While SIM7020G supports fewer frequency bands, it still covers most markets and is suitable for general IoT applications. Both modules are similar in network protocols and functional features, with extensive international and operator certifications.